

Entities and Relationships Summary:

1. tblusers (Users)

- Attributes:
 - user_id: INT, Primary Key, auto-increment.
 - first_name: VARCHAR(255), User's first name.
 - last_name: VARCHAR(255), User's last name.
 - username: VARCHAR(255), Unique username for logging in.
 - password: VARCHAR(255), Encrypted user password.
 - role: VARCHAR(50), Defines the role of the user (e.g., Admin, Customer, Manager).
- Relationships:
 - One-to-Many with `tblorders` (One user can place many orders)

2. tblorders (Orders)

- Attributes:
 - order_id: INT, Primary Key, auto-increment.
 - order_date: DATETIME, The date and time the order was placed.
 - shipper_id: INT, Foreign Key referencing tblshippers(shipper_id), defines which shipper will deliver the order.
 - user_id: INT, Foreign Key referencing tblusers(user_id), defines the user who placed the order.
 - amount: DECIMAL(10, 2), The total amount of the order.
 - status: VARCHAR(255), Status of the order (e.g., Pending, Delivered, Canceled).
- Relationships:
 - Many-to-One with `tblusers` (An order belongs to one user)
 - Many-to-One with `tblshippers` (An order is delivered by one shipper)
 - One-to-Many with `tblorderdetails` (An order can have many order details)

3. tblorderdetails (Order Details)

- Attributes:
 - order_id: INT, Foreign Key referencing tblorders(order_id), the order to which these details belong.
 - book_id: INT, Foreign Key referencing tblbooks(book_id), the book being ordered.
 - qty: INT, The number of copies of the book being ordered.
 - quoted_price: DECIMAL(10, 2), The price of the book at the time of the order (may differ from the regular price due to discounts or promotions).
- Relationships:
 - Many-to-One with `tblorders` (Each order detail belongs to one order)
 - Many-to-One with `tblbooks` (Each order detail is for one book)

4. tblbooks (Books)

- Attributes:
 - `book_id`: INT, Primary Key, auto-increment.
 - `isbn`: VARCHAR(13), Unique International Standard Book Number for the book.
 - `title`: VARCHAR(255), The title of the book.
 - `price`: DECIMAL(10, 2), The current price of the book.
 - `publisher_id`: INT, Foreign Key referencing `tblpublishers(publisher_id)`, defines the publisher of the book.
 - `stock_quantity`: INT, The number of copies available in stock.
- Relationships:
 - Many-to-One with `'tblpublishers'` (Each book is published by one publisher)
 - Many-to-Many with `'tblauthors'` via `'tblbooks_authors'` (A book can have multiple authors)
 - Many-to-Many with `'tblgenres'` via `'tblbooks_genres'` (A book can belong to multiple genres)
 - One-to-Many with `'tblorderdetails'` (A book can appear in multiple order details)
 - Many-to-Many with `tblpublishers` via `tblbooks_publishers`.

5. `tblauthors` (Authors)

- Attributes:
 - `author_id`: INT, Primary Key, auto-increment.
 - `first_name`: VARCHAR(255), Author's first name.
 - `last_name`: VARCHAR(255), Author's last name.
- Relationships:
 - Many-to-Many with `'tblbooks'` via `'tblbooks_authors'` (An author can write multiple books)

6. `tblbooks_authors` (Books–Authors Association)

- Attributes:
 - `book_id`: INT, Foreign Key referencing `tblbooks(book_id)`, the book written by the author.
 - `author_id`: INT, Foreign Key referencing `tblauthors(author_id)`, the author of the book.
- Relationships:
 - Many-to-One with `'tblbooks'` (Each record links a book to an author)
 - Many-to-One with `'tblauthors'` (Each record links an author to a book)

7. `tblgenres` (Genres)

- Attributes:
 - `genre_id`: INT, Primary Key, auto-increment.
 - `genre_name`: VARCHAR(255), The name of the genre (e.g., Fiction, Science Fiction, Romance).
- Relationships:

- Many-to-Many with `tblbooks` via `tblbooks\_genres` (A genre can apply to multiple books)

8. tblbooks_genres (Books-Genres Association)

- Attributes:
 - book_id: INT, Foreign Key referencing tblbooks(book_id), the book associated with the genre.
 - genre_id: INT, Foreign Key referencing tblgenres(genre_id), the genre of the book.
- Relationships:
 - Many-to-One with `tblbooks` (Each record links a book to a genre)
 - Many-to-One with `tblgenres` (Each record links a genre to a book)

9. tblpublishers (Publishers)

- Attributes:
 - publisher_id: INT, Primary Key, auto-increment.
 - publisher_name: VARCHAR(255), The name of the publisher.
 - address: VARCHAR(255), The address of the publisher.
- Relationships:
 - One-to-Many with `tblbooks` (A publisher can publish multiple books)
 - Many-to-Many with tblbooks via tblbooks_publishers.

10. tblshippers (Shippers)

- Attributes:
 - shipper_id: INT, Primary Key, auto-increment.
 - name: VARCHAR(255), The name of the shipping company or service.
 - phone_number: VARCHAR(20), The contact number for the shipper.
- Relationships:
 - One-to-Many with `tblorders` (A shipper can deliver multiple orders)

11. tblimports (Imports)

- Attributes:
 - import_id: INT, Primary Key, auto-increment.
 - supplier_name: VARCHAR(255), The name of the supplier providing the imported books.
 - import_date: DATETIME, The date and time the import occurred.
 - total_cost: DECIMAL(10, 2), The total cost of the import.
 - remark: VARCHAR(255), Any additional notes about the import.
- Relationships:
 - One-to-Many with `tblimports\_details` (Each import has multiple details)

12. tblimports_details (Import Details)

- Attributes:
 - import_id: INT, Foreign Key referencing tblimports(import_id), the import transaction to which the details belong.
 - book_id: INT, Foreign Key referencing tblbooks(book_id), the book being imported.
- Relationships:
 - Many-to-One with `tblimports` (Each record links an import to a book)
 - Many-to-One with `tblbooks` (Each record links a book to an import)

13. tblbooks_publishers (Books-Publishers Association)

- Attributes:
 - book_id: INT, Foreign Key referencing tblbooks(book_id)
 - publisher_id: INT, Foreign Key referencing tblpublishers(publisher_id)
 - published_date: DATE, The date when the book was published by this publisher (optional, if you need to track when each publisher published the book)
 - edition: VARCHAR(50), Optional, to indicate the edition or version of the book published by this publisher (e.g., 1st Edition, 2nd Edition)
- Relationships:
 - Many-to-One with tblbooks: Each record links a book to a publisher.
 - Many-to-One with tblpublishers: Each record links a publisher to a book.

14. tblexpenses (Expenses)

- Attributes:
 - expense_id: INT, Primary Key, auto-increment.
 - expense_type: VARCHAR(255), Describes the type of expense (e.g., Rent, Utilities, Salaries, Maintenance).
 - amount: DECIMAL(10, 2), The cost or value of the expense.
 - expense_date: DATE, The date the expense was recorded.
 - remarks: VARCHAR(255), Any additional notes or explanations regarding the expense.

Role of tblexpenses:

- Standalone Table: Since expenses are generally independent of other entities like orders, books, and customers, the tblexpenses table doesn't need foreign key relationships with other tables unless expenses are directly tied to specific operations (such as book purchases or staff-related expenses).

Summary of the ERD:

- ✚ Orders and Customers: A user can place many orders, and each order is handled by a specific shipper. The details of each order (books and quantities) are stored in the `'tblorderdetails'` table.
- ✚ Books and Authors: A book can have multiple authors, and an author can write multiple books. This is handled by the `'tblbooks_authors'` table, which represents the many-to-many relationship between books and authors.
- ✚ Books and Genres: A book can belong to multiple genres, and each genre can apply to multiple books. This many-to-many relationship is managed by the `'tblbooks_genres'` table.
- ✚ Imports: The system tracks book imports via the `'tblimports'` and `'tblimports_details'` tables, which associate imported books with specific import transactions.

This ERD is comprehensive and captures the core functionalities of a bookshop management system, covering key areas such as users, orders, books, authors, genres, publishers, and imports.